

Assignment9

Sema Tay

2025-12-16

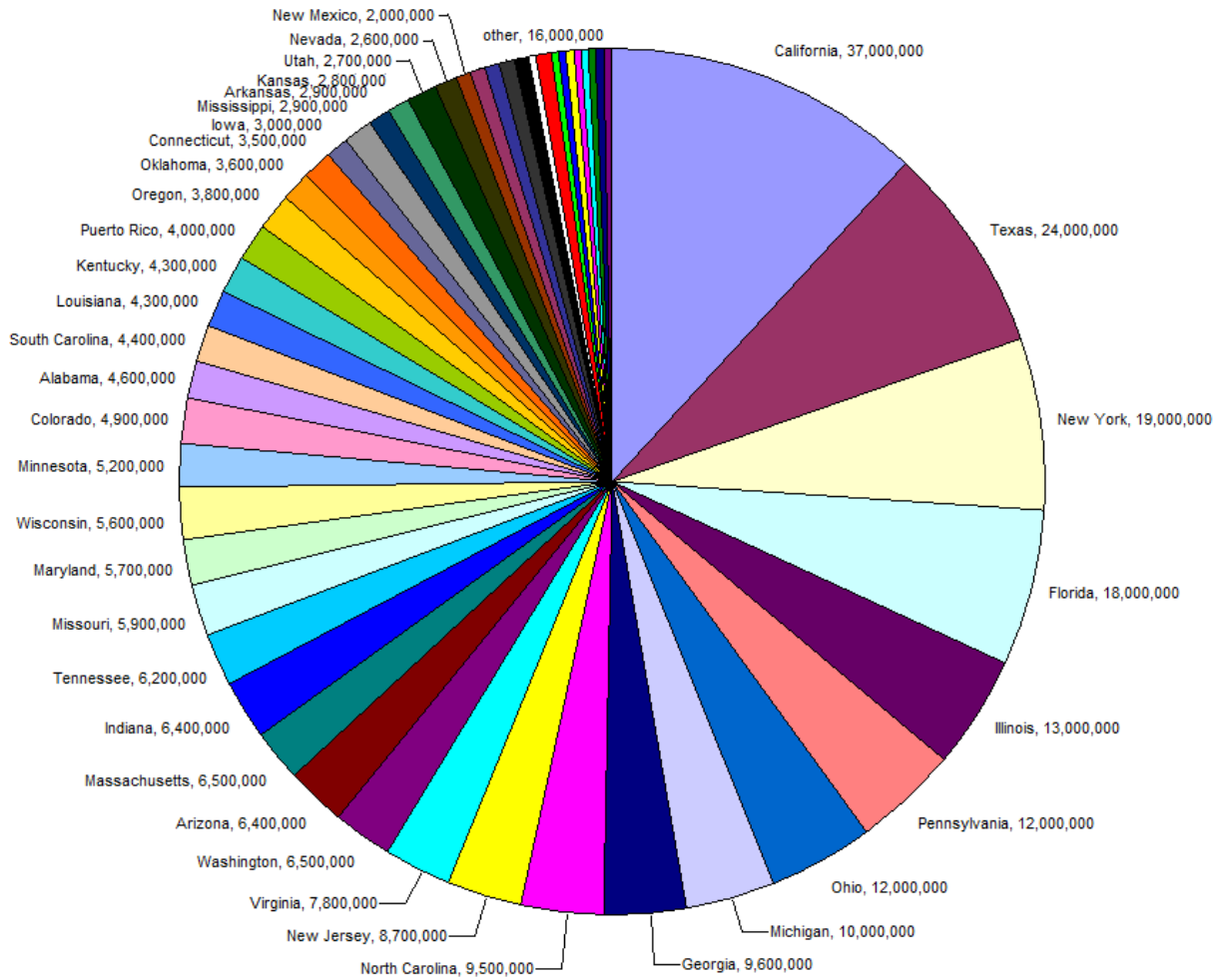
```
install.packages("magick")

## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)

library(magick)

## Linking to ImageMagick 6.9.10.23
## Enabled features: fontconfig, freetype, fftw, lcms, pango, webp, x11
## Disabled features: cairo, ghostscript, heic, raw, rsvg
## Using 16 threads

img <- "piechart.png"
photo <- image_read(img)
photo
```



This graph shows the distribution of the US population by state. Because there are so many categorical variables, a pie chart is not suitable for this data. Comparing states is difficult because the values are very close together. As the values decrease, it becomes harder to understand which part belongs to which state. The use of many colors that are very similar creates color clutter and makes reading difficult. Furthermore, we cannot obtain information about what percentage of the US population each state represents. A bar plot can be used instead of a pie chart for this data. This would make it easier to compare states. Using a bar plot also allows you to write the names of states that are marked as “other” in the graph and whose names cannot be written. The single-color nature of the bar plot also eliminates the color clutter found in a pie chart. Thus, we obtain a more readable and useful graph.

https://commons.wikimedia.org/wiki/File:Pie_chart_of_US_population_by_state.png